Actividad 6 - Estudio de la Transformada de Laplace

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1.

Las condiciones iniciales no son cero, lo que se complica métodos clásicos
La entrada del sistema es una función por tramos o una función especial, como una exponencial o un escalón unitario (función de Heaviside).

Ejemplo 1: Ecuación diferencial con condiciones iniciales no nulas

$$y'' + 3y' + 2y = 0 \quad y(0) = 1, \quad y'(0) = 0$$

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$$
$$\mathcal{L}\{y''\} = sY(s) - y(0)$$
$$\mathcal{L}\{y\} = Y(s)$$

$$s^2Y(s) - s(1) - 0 + 3(sY(s) - 1) + 2Y(s) = 0$$
$$(s^2 + 3s + 2)Y(s) - s - 3 = 0$$
$$Y(s) = \frac{s+3}{s^2 + 3s + 2}$$
$$\frac{s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$
$$s + 3 = A(s+2) + B(s+1)$$

- Para $s = -2$: $-2 + 3 = A(0) + B(-1) \rightarrow 1 = -B \rightarrow B = -1$
- Para $s = -1$: $-1 + 3 = A(1) + B(0) \rightarrow 2 = A \rightarrow A = 2$

$$Y(s) = \frac{2}{s+1} - \frac{1}{s+2}$$

$$y(t) = 2e^{-t} - e^{-2t}$$

Ejemplo 2: Ecuación con función escalón

$$y'' + y = u_\pi(t), \quad y(0) = 0, \qquad y'(0) = 0$$
$$\mathcal{L}\{y''\} + \mathcal{L}\{y\} = \mathcal{L}\{u_\pi(t)\}$$
$$s^2Y(s) + Y(s) = \frac{e^{-\pi s}}{s} = Y(s) = \frac{e^{-\pi s}}{s(s^2+1)}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = \int_0^t \sin(\tau)\, d\tau = 1 - \cos(t)$$
$$y(t) = u_\pi(t) * [1 - \cos(t - \pi)]$$

2.

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt$$

- $f(t) = (1 + e^{2t})^2$

$$f(t) = (1 + e^{2t})^2 = 1 + 2e^{2t} + e^{4t}$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{1\} + 2\mathcal{L}\{e^{2t}\} + \mathcal{L}\{e^{4t}\}$$

$\mathcal{L}\{1\} = \frac{1}{s}$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \; para \; s > a$$

$$\mathcal{L}\{f(t)\} = \frac{1}{s} + \frac{2}{s-2} + \frac{1}{s-4}$$

- $f(t) = 4e^{-3t}$

$$\mathcal{L}\{f(t)\} = 4 * \mathcal{L}\{e^{-3t}\} = 4 * \frac{1}{s+3} = \frac{4}{s+3}$$

- $f(t) = t^4 - 3t^2 + 9$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$
$$\mathcal{L}\{t^4\} = \frac{4!}{s^5} = \frac{24}{s^5}$$
$$\mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}$$
$$\mathcal{L}\{9\} = \frac{9}{s}$$

$$\mathcal{L}\{f(t)\} = \frac{24}{s^5} - \frac{6}{s^3} + \frac{9}{s}$$

- $f(t) = e^{2t}\cos(3t)$

$$\mathcal{L}\{e^{at}\cos(bt)\} = \frac{s-a}{(s-a)^2 + b^2}$$
$$\mathcal{L}\{f(t)\} = \frac{s-2}{(s-2)^2 + 9}$$

3.

- $f(s) = \frac{3s-15}{2s^2-45s+10}$

$$D = (-45)^2 - 4(2)(10) = 2025 - 80 = 1945$$
$$\mathcal{L}^{-1}\left\{\frac{s-a}{(s-a)^2 + b^2}\right\} = e^{at}\cos(bt)$$

- $f(s) = \frac{6}{(s-1)^4}$

$$\mathcal{L}^{-1}\left\{\frac{n!}{(s-a)^{n+1}}\right\} = t^n e^{at}$$
$$\frac{6}{(s-1)^4} = \frac{3!}{(s-1)^4} \rightarrow n = 3, a = 1$$
$$\mathcal{L}^{-1}\left\{\frac{6}{(s-1)^4}\right\} = t^3 e^t$$

- $f(s) = \frac{1}{s^2} - \frac{1}{s+1}$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$$
$$\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$$
$$f(t) = t - e^{-t}$$

- $f(s) = \frac{(s+2)^2}{s^3}$

$$(s+2)^2 = s^2 + 4s + 4$$
$$f(s) = \frac{s^2 + 4s + 4}{s^3} = \frac{1}{s} + \frac{4}{s^2} + 4$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$
$$\mathcal{L}^{1}\left\{\frac{4}{s^2}\right\} = 4t$$
$$\mathcal{L}^{1}\left\{\frac{4}{s^3}\right\} = 2t^2$$

$$f(t) = 1 + 4t + 2t^2$$

4.

- $y'' + y = \sin(3t)\,, \quad y(0) = 0,\ y'(0) = 0$

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s)$$
$$L\{y\} = Y(s)$$
$$L\{sint(3t)\} = \frac{3}{s^2 + 9}$$

$$s^2Y(s) + Y(s) = \frac{3}{s^2 + 9} \rightarrow Y(s)(s^2 + 1) = \frac{3}{s^2 + 9}$$
$$Y(s) = \frac{3}{(s^2 + 9)(s^2 + 1)}$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{3}{(s^2 + 1)(s^2 + 9)}\right\} = 3\int_0^t \sin(\tau) * \frac{\sin[3(t - r)]}{3}\,dr$$
$$= \int_0^t \sin\,(r)\sin[3(t - r)]\,dr$$

$$Y(s) = \frac{3}{(s^2 + 1)(s^2 + 9)}$$

- $2y'' + y' - y = 0,\ y(0) = 1,\ y'(0) = 4$

$$2[s^2Y(s) - s(1) - 4] + [sY(s) - 1] - Y(s) = 0$$
$$2[s^2Y(s) - 2s - 8 + sY(s) - 1 - Y(s) = 0 \rightarrow (2s^2 + s - 1)Y(s) = 2s + 9$$

$$Y(s) = \frac{2s + 9}{2s^2 + s - 1}$$
$$2s^2 + s - 1 = (2s - 1)(s + 1)$$
$$\frac{2s + 9}{(2s - 1)(s + 1)} = \frac{A}{2s - 1} + \frac{B}{s + 1}$$
$$2s + 9 = A(s + 1) + B(2s - 1)$$

$$Y(s) = \frac{20}{3(2s - 1)} - \frac{7}{3(s + 1)}$$
$$\mathcal{L}^{-1}\left\{\frac{1}{s - a}\right\} = e^{at}$$
$$y(t) = \frac{20}{3}e^{\frac{1}{2}t} - \frac{7}{3}e^{-t}$$

- $y'' - 2y' + 5y = 0,\ y(0) = 2.\ y'(0) = 4$

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$$
$$\mathcal{L}\{y'\} = sY(s) - y(0)$$

$$s^2Y(s) - 2s - 4 - 2(sY(s) - 2) + 5Y(s) = 0$$
$$s^2Y(s) - 2s - 4 - 2sY(s) + 4 + 5Y(s) = 0$$
$$(s^2 - 2s + 5)Y(s) = 2s \rightarrow Y(s) = \frac{2s}{s^2 - 2s + 5}$$

$$s^2 - 2s + 5 = (s - 1)^2 + 2^2 \rightarrow Y(s) = \frac{2(s - 1 + 1)}{(s - 1)^2 + 4}$$
$$= \frac{2(s - 1)}{(s - 1)^2 + 4} + \frac{2}{(s - 1)^2 + 4}$$

$$y(t) = 2e^t cos(2t) + e^t \sin(2t)$$

- $w'' + w = t^2 + 2, \quad w(0) = 1, \; w'(0) = -1$

$$\mathcal{L}\{w''\} = s^2W(s) - sw(0) - w'(0) = s^2W(s) - s - (-1)$$
$$= s^2W(s) - s + 1$$

$$\mathcal{L}\{w\} = W(s), \qquad \mathcal{L}\{t^2 + 2\} = \frac{2}{s^3} + \frac{2}{s}$$

$$s^2W(s) - s + 1 + W(s) = \frac{2}{s^3} + \frac{2}{s} \rightarrow (s^2 + 1)W(s) = \frac{2}{s^3} + \frac{2}{s} + s - 1$$

$$W(s) = \frac{\frac{2}{s^3} + \frac{2}{s} + s - 1}{s^2 + 1}$$

$$w(t) = \frac{t^2}{s^2 + 1}$$

- $z'' + 5z' - 6z = 21e^t, \; z(0) = -1, \; z'(0) = 9$

$$\mathcal{L}\{z''\} = s^2Z(s) - sz(0) - z'(0) = s^2Z(s) + s - 9$$
$$\mathcal{L}\{z'\} = sZ(s) - z(0) = sZ(s) + 1$$
$$\mathcal{L}\{z\} = Z(s), \qquad \mathcal{L}\{21e^t\} = \frac{21}{s - 1}$$

$$s^2Z(s) + s - 9 + 5(sZ(s) + 1) - 6Z(s) = \frac{21}{s - 1}$$
$$s^2Z(s) + 5sZ(s) - 6Z(s) + s - 9 + 5$$
$$(s^2 + 5s - 6)\, Z(s) = \frac{21}{s - 1} - s + 4$$

$$s^2 + 5s + 6 = (s + 6)(s - 1)$$

$$Z(s) = \frac{21}{(s - 1)(s + 6)} - \frac{s - 4}{(s - 1)(s + 6)} \rightarrow Z(s) = \frac{17}{(s - 1)(s + 6)}$$

$$\frac{17}{(s - 1)(s + 6)} = \frac{A}{s - 1} + \frac{B}{s + 6}$$

$$\frac{17}{(s-1)(s+6)} = \frac{A}{s-1} + \frac{B}{s+6}$$

$$17 = A(s+6) + B(s-1)$$

$$z(t) = \frac{17}{7}e^{t} - \frac{17}{7}e^{-6t}$$